ELECTRODYNAMICS II · SEMESTER 5

Induction and Maxwell equations

Module 2 · Prepare

Start with the idea. Learn the mathematics. Work the questions. Check the result.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

How to use this book

P means prerequisite, M means mathematics, T means topic and Q means the numbered practice question. Topic codes restart in each module. Read P1 and M1 first, then follow the topics in order.

REMIND ME

A fact from school that we will use again.

MATHS TOOL

The meaning of a symbol and the operation it asks you to do.

THE IDEA

A physical explanation before the equation.

RESULT

The statement to carry into an exam.

EXAM TRAP

An assumption, sign or unit that changes the answer.

CHECK YOURSELF

Cover the answer and try. Then explain why your answer makes sense.

One rule for this module

> **KEEP THIS IN MIND**
>
> Changing fields and moving conductors can drive charge; Maxwell’s equations keep that motion consistent with conservation of charge.

The whole module in five sentences

1. An electric field drives current through a resistive conductor.
2. A source supplies energy per charge, and motion through a magnetic field can produce an emf.
3. Changing magnetic flux produces circulation of E, even without a wire.
4. Inductance measures flux linkage per current, and a magnetic field stores energy.
5. Maxwell adds displacement current so that changing electric fields also produce magnetic circulation.

Syllabus route

Units 6–8 · 12 hours · 18 theory marks. Griffiths, 5th ed., §§7.1.1–7.1.3, 7.2.1–7.2.4, 7.3.1–7.3.3 and 7.3.5–7.3.6.

Definitions and derivations are independently explained from the named textbook sections. Worked numerical values are original practice unless a past-paper tag is printed.

Topic route

Unit 6 → T1–T3; unit 7 → T4–T7; unit 8 → T8–T11.

T1 · Ohm’s law in a material
T2 · Electromotive force
T3 · Motional emf and energy supply
T4 · Faraday’s law and Lenz’s rule
T5 · Induced electric field
T6 · Self and mutual inductance
T7 · Magnetic field energy
T8 · Charge conservation and Maxwell’s repair
T9 · The four Maxwell equations
T10 · Maxwell equations in matter
T11 · Electromagnetic boundary conditions

P1 School physics bridge

Current through an area

$I = dQ/dt$ is measured in amperes. Current density $\mathbf{J}$ tells how much current flows through each unit cross-sectional area: $I = \int \mathbf{J} \cdot d\mathbf{a}$. For uniform perpendicular flow, $I = JA$. A wire of resistance R obeys $V = IR$ under ohmic conditions and dissipates $P = VI = I^2R$.

Fields and forces

A positive charge q experiences $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. E has units V/m, B tesla, v m/s. The cross product points along the right-hand thumb when fingers turn from v toward B. Magnetic force is perpendicular to the charge's velocity, so it does no instantaneous work on that charge.

Flux and sources

Magnetic flux through a flat area A in uniform B is $\Phi_B = BA\cos\theta$, in webers (Wb); θ is the angle between B and the **area normal**. For a coil of N turns each linking the same flux, linkage is $N\Phi_B$. A source emf is energy supplied per charge, measured in volts; despite its name it is not a force in newtons.

Material fields

$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P}$ and $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$. P and M describe polarization and magnetization. Subscript f denotes free charge or current. In a linear isotropic medium, $D = \epsilon E$ and $B = \mu H$.

M1 Mathematics from the beginning

Slope with respect to time

The derivative $d\Phi/dt$ means flux change per second at an instant. If $\Phi = at^2$, the rate is $2at$. If flux falls from 0.020 to 0.005 Wb in 0.030 s at a constant rate, $d\Phi/dt = -0.50$ Wb/s. Faraday's minus sign reverses this signed rate.

Line and surface sums

$\oint \mathbf{E} \cdot d\mathbf{l}$ adds the component of E along each short piece of a closed path. A circle in a tangential field of constant magnitude gives $E(2\pi r)$. An area integral $\int \mathbf{B} \cdot d\mathbf{a}$ adds normal field × tiny area. Match the path and area signs using the right-hand rule.

Divergence and curl

Divergence measures net outward flow per tiny volume. Its integral over a volume is the outward flux through the boundary (divergence theorem). Curl measures circulation per tiny area. Its flux through a surface equals circulation along the boundary (Stokes' theorem). Thus $\oint E \cdot dl = -d\int B \cdot da/dt$ becomes $\nabla \times E = -\partial B/\partial t$ for a fixed surface.

Why divergence of curl is zero

Interior edges cancel when the circulations of neighbouring tiny faces are added. A closed surface has no outer edge left. Symbolically, $\nabla \cdot$

$(\nabla \times B) = 0$. This identity is the key check of charge conservation in Ampère's law.

Integrating a changing current

If an inductor needs voltage $v_L = L\,dI/dt$, power supplied is $p = v_L I$. Energy added in time dt is $dU = p\,dt = LI\,dI$. Summing from zero to I gives $U = L\int_0^I I'\,dI' = LI^2/2$. The prime labels a temporary integration variable.

Jump conditions

At an interface define normal n from region 1 to 2. A pillbox tests normal components. A narrow rectangular loop tests tangential components. Terms proportional to shrinking area vanish if the fields stay finite. An actual surface charge or surface current survives the limit.

T1 Ohm’s law in a material

> Purpose: Current density responds to the local electric driving field.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Current density responds to the local electric driving field.

1. For a straight uniform wire, $J = I/A$ and $E = V/\ell$.
   Why: Uniform cross-section and steady flow allow constant averages.
2. Substitute into $J = \sigma E$ and rearrange.
   Why: The same local response applies throughout the wire.
3. Power converted to heat per volume is $\mathbf{J} \cdot \mathbf{E}$.
   Why: Field work per charge times charge flow gives energy transfer.

RESULT

$\mathbf{J} = \sigma \mathbf{E}$ for an ohmic conductor at rest; $R = \ell/(\sigma A)$.

EXAM TRAP

Conductivity σ is not a surface charge density in this topic; its unit is S/m.

CHECK YOURSELF

If length doubles at fixed area and conductivity, what happens to R?
Answer: R doubles.

This topic supports Q1, Q2, Q21.

Work it through · Q1

Skill: Microscopic Ohm law. Model · syllabus-based

State local Ohm's law, define conductivity and distinguish it from resistivity.

For an isotropic conductor at rest, $\mathbf{J} = \sigma \mathbf{E}$. Conductivity σ has units S/m; resistivity is $\varrho = 1/\sigma$ in Ω m. It is an empirical linear response, not a universal law for all materials.

CHECK THE ENDPOINT

$J = \sigma E$; $\varrho = 1/\sigma$.

Work it through · Q2

Skill: Wire resistance scaling. Model · syllabus-based

Derive the resistance of a uniform wire of length ℓ and area A from local Ohm's law.

Use $J = I/A$ and $E = V/\ell$. Then $I/A = \sigma V/\ell$, giving $V/I = \ell/(\sigma A)$. The assumptions are uniform conductivity, cross-section and steady current.

CHECK THE ENDPOINT

$R = \ell/(\sigma A)$.

Work it through · Q21

Skill: Conduction and Joule power. Model · syllabus-based

A wire has ℓ=2 m, A=1.0×10^{-6} m² and σ=5.0×10^{7} S/m. Apply 0.10 V. Find R, I, E, J and total power.

1. $R = \ell/(\sigma A) = 0.040\ \Omega$.
   Why: The wire is uniform and ohmic.

2. $I = V/R = 2.5$ A and $E = V/\ell = 0.050$ V/m.
   Why: Use the terminal voltage along its length.

3. $J = \sigma E = 2.5 \times 10^6$ A/m², also I/A.
   Why: Local and global current descriptions must agree.

4. $P = VI = 0.25$ W; equivalently $JE(A\ell) = 0.25$ W.
   Why: Field power density integrates to the circuit power.

CHECK THE ENDPOINT

$R = 0.040\ \Omega$, I=2.5 A, E=0.050 V/m, $J = 2.5 \times 10^6$ A/m², P=0.25 W.

T2 Electromotive force

Purpose: An emf accounts for energy supplied around a circuit per unit charge.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

An emf accounts for energy supplied around a circuit per unit charge.

1. A battery's chemical source pushes charge against an electrostatic potential rise.
   Why: Electrostatic forces alone have zero closed-loop work.
2. In a moving circuit the magnetic part $\mathbf{u} \times \mathbf{B}$ can contribute to the circuit emf.
   Why: u is the velocity of the conductor element.
3. An emf need not equal the terminal voltage of a source delivering current through internal resistance.
   Why: Some source energy is dissipated internally.

RESULT

$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l}$ where f is driving force per unit charge.

EXAM TRAP

The symbol emf denotes volts, not newtons.

CHECK YOURSELF

A battery has emf 12 V, internal resistance 1 Ω and supplies 2 A. Terminal voltage?
Answer: 10 V.

This topic supports Q3, Q4, Q22.

Work it through · Q3

Skill: Emf meaning and units. Model · syllabus-based

Define emf and explain why its name can be misleading.

Emf is closed-path driving work per unit charge, $\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l}$. It has units J/C = volt. It is not a mechanical force and need not equal the loaded source's terminal voltage.

CHECK THE ENDPOINT

Emf measures energy per charge, in volts.

Work it through · Q4

Skill: Electrostatic circulation. Model · syllabus-based

Can a purely electrostatic field provide a nonzero emf around a closed loop? Explain.

No. $\mathbf{E} = -\nabla V$ for a single-valued electrostatic potential, so $\oint E \cdot dl = 0$. A battery needs a non-electrostatic source force; an induced electric field is not electrostatic.

CHECK THE ENDPOINT

Purely electrostatic closed-loop emf is zero.

Work it through · Q22

Skill: Battery terminal behaviour. Model · syllabus-based

A 12 V source of internal resistance 1 Ω drives a 5 Ω load. Find current, terminal voltage and the power budget.

1. The closed-loop equation is $12 = I(1 + 5)$.
   Why: Both resistances carry the same current.

2. I=2 A and $V_{\text{term}} = 12 - Ir = 10$ V.
   Why: Internal voltage drop reduces the external voltage.

3. Source power is $\mathcal{E}I = 24$ W; internal heating is $I^2r = 4$ W and load power is $I^2R = 20$ W.
   Why: Energy supplied equals total dissipation.

CHECK THE ENDPOINT

I=2 A, terminal voltage 10 V; 24 W supplied = 4 W internal + 20 W load.

T3 Motional emf and energy supply

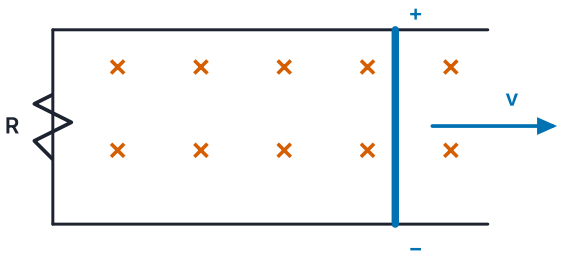
> Purpose: A moving conductor cuts magnetic flux and separates charge.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

A moving conductor cuts magnetic flux and separates charge.



B into page (×); rod moves right

With velocity right and B into the page, v × B points up: the top of the rod becomes positive.

1. Charges moving with the rod feel $q\mathbf{v} \times \mathbf{B}$.
   Why: Lorentz force separates signs along the rod.
2. The emf along it has magnitude $B\ell v$.
   Why: Integrate the force per charge along its length.
3. A closed resistive circuit carries $I = B\ell v/R$ and has magnetic drag $F = BI\ell$.
   Why: The induced current opposes the increasing flux.
4. Mechanical power Fv equals I^2R in steady motion.
   Why: The pulling agent supplies the electrical energy.

RESULT

$\mathcal{E} = B\ell v$ for mutually perpendicular B, rod length and velocity.

EXAM TRAP

The magnetic field itself does no work on individual charges; the external mechanical agent supplies the heat.

CHECK YOURSELF

Double v in the same circuit. How does heating change?

Answer: I doubles, so I^2R quadruples.

This topic supports Q5, Q6, Q23.

Work it through · Q5

Skill: Rod voltage. Model · syllabus-based

A 0.40 m rod moves at 3.0 m/s perpendicular to a 0.50 T field and to its own length. Find the emf magnitude.

$\mathcal{E} = B\ell v = (0.50)(0.40)(3.0) = 0.60$ V. Charge separation is along $\mathbf{v} \times \mathbf{B}$; the positive end depends on the stated vector directions.

CHECK THE ENDPOINT

$\mathcal{E} = 0.60$ V.

Work it through · Q6

Skill: Motional energy source. Model · syllabus-based

If magnetic forces do no work on individual charges, where does the heat in a moving-rod circuit come from?

The pulling agent works against magnetic drag on the current-carrying rod. In steady motion its mechanical power equals the resistor's heating power. Magnetic forces redirect motion but supply no net work on a charge.

CHECK THE ENDPOINT

The external mechanical agent supplies the energy.

Work it through · Q23

Skill: Moving rod power balance. Model · syllabus-based

The rod in Q5 completes a circuit of total resistance 2 Ω. Find current, magnetic drag and pulling power at constant speed.

1. $I = \mathcal{E}/R = 0.60/2 = 0.30$ A.
   Why: The circuit is resistive and induction transients are neglected.

2. $F = BI\ell = (0.50)(0.30)(0.40) = 0.060$ N opposite motion.
   Why: Lenz's law fixes the drag direction.

3. $P_{\text{pull}} = Fv = 0.18$ W and $I^2R = 0.18$ W.
   Why: Steady motion leaves no kinetic-energy increase.

CHECK THE ENDPOINT

I=0.30 A, drag=0.060 N, input power=0.18 W equals heating.

T4 Faraday’s law and Lenz’s rule

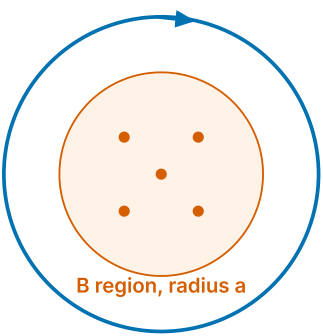
> Purpose: Changing magnetic flux creates an emf that opposes the flux change.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Changing magnetic flux creates an emf that opposes the flux change.



Increasing B out of page → clockwise E

Dots mean B toward the reader. The surrounding induced E circulation opposes the increasing outward flux; it persists outside the B region.

1. Choose the positive area normal and path orientation first.

   Why: The flux and emf signs must be linked.

2. Calculate changes in B, area or angle, not only B.
   Why: Any of these changes can alter flux.

3. Apply the minus sign and infer the current direction.
   Why: The induced field opposes the change, not necessarily the original field.

RESULT

$\mathcal{E} = -d\Phi_B/dt$ for a loop; $\mathcal{E} = -N\, d\Phi_B/dt$ for N identical turns.

EXAM TRAP

Flux can be zero at an instant while its derivative and emf are nonzero.

CHECK YOURSELF

A positive flux is increasing. Is the emf positive or negative in the matching orientation?
Answer: Negative.

This topic supports Q7, Q8, Q24, Q25.

Work it through · Q7

Skill: Lenz sign. Model · syllabus-based

State Lenz's rule and distinguish opposition to flux from opposition to flux change.

The induced current produces magnetic effects opposing the change in linked flux. If an existing flux decreases, the induced field acts in the same

direction as that flux to oppose its decrease.

CHECK THE ENDPOINT

The opposition is to change, not always to the field itself.

Work it through · Q8

Skill: Changing area and angle. Model · syllabus-based

List three ways to change the flux through a flat loop in uniform B.

Since $\Phi_B = BA\cos\theta$, changing B, loop area A or orientation θ can change flux. θ is measured from the area normal. A nonzero flux does not by itself imply an emf.

CHECK THE ENDPOINT

Emf requires a time derivative of linked flux.

Work it through · Q24

Skill: Flux ramp in a coil. Model · syllabus-based

A 200-turn coil of area 0.010 m² has its normal along a field that rises uniformly from 0.10 T to 0.40 T in 0.050 s. Find its emf, including sign.

1. Choose positive normal along B, so $\Phi_B = BA$.
   Why: The angle is zero.
2. $dB/dt = 0.30/0.050 = 6.0$ T/s.
   Why: The ramp is uniform.
3. $\mathcal{E} = -NA\,dB/dt = -200(0.010)(6.0) = -12$ V.
   Why: Use total linkage rather than one-turn flux.

CHECK THE ENDPOINT

−12 V in the positive-loop orientation; magnitude 12 V.

Work it through · Q25

Skill: Rotating flux differentiation. Model · syllabus-based

A coil has N turns and area A and rotates with angular speed ω in uniform B. Derive its sinusoidal emf when its normal initially points along B.

1. The angle is $\theta = \omega t$, giving linkage $\lambda = NAB\cos\omega t$.
   Why: Rotation changes orientation, not B or A.
2. Differentiate: $d\lambda/dt = -NAB\omega\sin\omega t$.
   Why: The chain rule contributes ω.
3. Faraday's minus sign gives $\mathcal{E} = NAB\omega\sin\omega t$.
   Why: The emf opposes the signed flux change.
4. At maximum linkage the emf is zero; at zero linkage its magnitude is maximum.
   Why: A slope is not the same as the function value.

CHECK THE ENDPOINT

Peak emf $\mathcal{E}_0 = NAB\omega$.

T5 Induced electric field

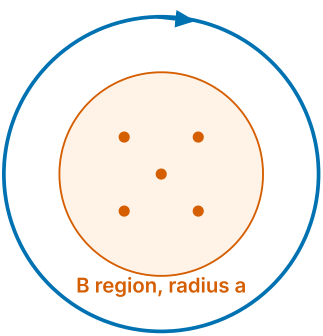
> Purpose: A changing B creates circulating E even where no conducting wire is present.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

A changing B creates circulating E even where no conducting wire is present.



Increasing B out of page → clockwise E

Dots mean B toward the reader. The surrounding induced E circulation opposes the increasing outward flux; it persists outside the B region.

1. For an axially symmetric changing B, choose a circular path.
   Why: Symmetry makes E tangential and equal along the circle.

2. Inside a uniform-field disk, enclosed flux is $B\pi r^2$.
   Why: Only the area inside the path contributes.

3. Outside a disk of radius a, enclosed flux is $B\pi a^2$.
   Why: B is confined to the disk, so added area contributes no flux.

4. Divide the flux derivative by $2\pi r$ to find signed E_ϕ.
   Why: The line integral is E times the circumference.

RESULT

$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$; $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$.

EXAM TRAP

An induced E generally cannot be written as only $-\nabla V$ with a single-valued scalar V.

CHECK YOURSELF

Can a test charge experience induced E outside the region containing B?
Answer: Yes. The enclosed changing flux gives nonzero circulation.

This topic supports Q9, Q10, Q26, Q27, Q38.

Work it through · Q9

Skill: Induced field without wire. Model · syllabus-based

Can there be induced E at a location containing no conductor? Give the governing law.

Yes. $\nabla \times E = -\partial B/\partial t$ describes a field in space. A wire merely provides mobile charges that respond; the induced E exists without it.

CHECK THE ENDPOINT

Changing magnetic flux produces electric circulation without a wire.

Work it through · Q10

Skill: Potential limitation. Model · syllabus-based

Why cannot a general induced E be described using only a single-valued scalar electrostatic potential?

A gradient has zero curl and zero closed-loop integral. A changing B gives $\oint E \cdot dl = -d\Phi_B/dt$, which can be nonzero. Thus $E = -\nabla V$ alone is insufficient.

CHECK THE ENDPOINT

Nonzero circulation rules out a purely scalar electrostatic description.

Work it through · Q26

Skill: Electric field inside changing-flux disk. Model · syllabus-based

Uniform $B(t)\hat{z}$ occupies a circular region of radius a. Derive the azimuthal E for r<a, with positive circulation counterclockwise viewed from +z.

1. Choose a concentric circle of radius r.
   Why: Axial symmetry makes E tangential and constant.
2. The enclosed flux is $B\pi r^2$.
   Why: The whole surface lies within the field region.

3. $E_\phi(2\pi r) = -\pi r^2 \dot{B}$.
   Why: Apply Faraday's law with the right-hand normal +z.

4. Thus $E_\phi = -(r/2)\dot{B}$.
   Why: Divide by circumference.

CHECK THE ENDPOINT

Inside: $\mathbf{E} = -(r/2)\dot{B}\,\hat{\boldsymbol{\phi}}$.

Work it through · Q27

Skill: Electric field outside changing-flux disk. Model · syllabus-based

For Q26 find E at r>a and evaluate its magnitude for a=0.10 m, r=0.20 m and $\dot{B} = 4$ T/s.

1. Only the disk contributes flux, so $\Phi = B\pi a^2$.
   Why: B is zero outside the disk.

2. $2\pi r E_\phi = -\pi a^2 \dot{B}$, hence $E_\phi = -a^2\dot{B}/(2r)$.
   Why: The path circumference still uses r.

3. The magnitude is $(0.10)^2(4)/(2 \times 0.20) = 0.10$ V/m.
   Why: Substitute in SI units.

4. The inside and outside formulas agree at r=a.
   Why: This checks continuity at the boundary.

CHECK THE ENDPOINT

Outside: $E_\phi = -a^2\dot{B}/(2r)$; magnitude 0.10 V/m here.

Work it through · Q38

Skill: Unifying stationary and motional induction. Model · syllabus-based

Compare induction in a stationary loop with induction in a moving conductor, including the correct emf expression, sign and energy source.

1. For a stationary loop, $\mathcal{E} = \oint E \cdot dl = -d\Phi_B/dt$.
   Why: Only the electric force contributes along a fixed path.

2. For a moving material loop, $\mathcal{E} = \oint (E + u \times B) \cdot dl$, where u is the conductor velocity.
   Why: Charges share the conductor's transport motion.

3. For an ordinary moving closed circuit, the combined expression equals the negative rate of linked flux.
   Why: Both field variation and changing surface geometry enter the flux derivative.

4. The loop orientation and area normal must be paired.
   Why: Faraday's sign is meaningful only with this convention.

5. Lenz's rule opposes the flux change. In a moving rod this creates drag, and the pulling work becomes electrical heating.
   Why: Energy conservation rules out self-amplifying motion.

6. In a stationary coil near a changing source field, the field-generating apparatus supplies the transferred energy.
   Why: A nonconservative E can transfer energy without mechanical motion of the receiving loop.

CHECK THE ENDPOINT

Stationary induction uses circulating E; motional induction includes $u \times B$; both obey the oriented flux rule under the stated conditions.

T6 Self and mutual inductance

Purpose: Inductance measures how strongly circuit current links magnetic flux.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Inductance measures how strongly circuit current links magnetic flux.

1. In a fixed linear geometry, magnetic field is proportional to current.
   Why: Superposition applies.

2. Flux linkage therefore scales with current.
   Why: The coil area and turns are fixed.

3. Differentiate the linkage and use Faraday's law.
   Why: A changing current induces an opposing emf.

4. Mutual inductance links two circuits; reciprocity gives $M_{12} = M_{21}$ for ordinary reciprocal linear media.
   Why: The mutual magnetic energy is symmetric.

RESULT

$N\Phi = LI$; $\mathcal{E}_L = -L\,dI/dt$ for constant L; $N_2\Phi_{21} = MI_1$.

EXAM TRAP

L is a circuit property with units henry, not a resistance. The negative sign belongs to induced emf; passive voltage drop uses $+L\,dI/dt$.

CHECK YOURSELF

What current change induces 1 V in 0.20 H?
Answer: A rate magnitude of 5 A/s.

This topic supports Q11, Q12, Q13, Q28, Q29.

Work it through · Q11

Skill: Self-inductance definition. Model · syllabus-based

Define self-inductance and the henry, stating the linearity assumption.

For a fixed linear circuit, flux linkage is $N\Phi = LI$. One henry corresponds to one weber-turn per ampere, or an induced emf magnitude of one volt when current changes at one ampere per second.

CHECK THE ENDPOINT

$L = N\Phi/I$ in henries for a linear circuit.

Work it through · Q12

Skill: Back emf versus terminal drop. Model · syllabus-based

Distinguish the induced emf of an ideal inductor from its passive terminal voltage drop.

The induced emf opposes the current change: $\mathcal{E}_L = -L\,dI/dt$. With current entering the positive-labelled terminal, the passive drop is $v_L = +L\,dI/dt$. They describe the same induction with different sign conventions.

CHECK THE ENDPOINT

Back emf is negative relative to increasing current; passive drop is positive.

Work it through · Q13

Skill: Mutual-inductance direction convention. Model · syllabus-based

Define mutual inductance and explain why coil orientation matters.

Linkage in coil 2 due to I_1 is $\lambda_{21} = MI_1$. Then $\mathcal{E}_2 = -M\,dI_1/dt$. Reversing one coil's reference normal changes the signed linkage and M, while its coupling magnitude is unchanged.

CHECK THE ENDPOINT

State winding directions or use a consistent dot convention.

Work it through · Q28

Skill: Solenoid self-inductance. Model · syllabus-based

Derive L for a long solenoid with N turns, length ℓ, area A and uniform linear permeability μ.

1. Inside, $H = NI/\ell$ and $B = \mu NI/\ell$.
   Why: Neglect end effects and use the free-current circulation law.
2. Each turn links $\Phi = BA$.
   Why: The field is approximately uniform over the cross-section.
3. Total linkage is $N\Phi = \mu N^2 AI/\ell$.
   Why: N identical turns link the same flux.
4. Divide linkage by current.
   Why: Inductance is defined by $\lambda = LI$.

CHECK THE ENDPOINT

$L = \mu N^2 A/\ell$.

Work it through · Q29

Skill: Coupled coils calculation. Model · syllabus-based

Two ideal coaxial long coils share area 2×10^{-4} m² and length 0.20 m, with 500 and 1000 turns. In air, find M and the emf magnitude in coil 2 when $dI_1/dt = 3$ A/s.

1. Coil 1 makes $B_1 = \mu_0 N_1 I_1 / \ell$.
   Why: Assume all of its uniform interior flux links coil 2.

2. $\lambda_{21} = N_2 B_1 A = M I_1$, so $M = \mu_0 N_1 N_2 A / \ell$.
   Why: Use the common cross-sectional area.

3. $M = 6.283 \times 10^{-4}$ H.
   Why: Substitute the specified geometry.

4. $|\mathcal{E}_2| = M|\dot{I}_1| = 1.885 \times 10^{-3}$ V.
   Why: Faraday's law differentiates linkage.

CHECK THE ENDPOINT

M=0.6283 mH; induced emf magnitude 1.885 mV.

T7 Magnetic field energy

> Purpose: Building current requires work, stored in the magnetic field.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Building current requires work, stored in the magnetic field.

1. The source must supply a voltage opposing the back emf.
   Why: Otherwise current cannot increase.
2. Multiply $L\,dI/dt$ by current to get input power.
   Why: Power is voltage times current.
3. Integrate $LI\,dI$ from zero to the final current.
   Why: Energy is accumulated power.
4. For a solenoid insert its field and geometry to identify energy per volume.
   Why: The uniform-field limit connects circuit and field descriptions.

RESULT

$U = LI^2/2$; in a linear nondispersive medium $u_B = \mathbf{B} \cdot \mathbf{H}/2$. In vacuum $u_B = B^2/(2\mu_0)$.

EXAM TRAP

In a nonlinear or hysteretic medium, blindly using $BH/2$ can fail; incremental work depends on the path.

CHECK YOURSELF

If current doubles in an ideal linear inductor, how does U change?
Answer: It quadruples.

This topic supports Q14, Q30, Q31, Q39.

Work it through · Q14

Skill: Inductor energy calculation. Model · syllabus-based

Find the energy of a 0.30 H inductor carrying 2.0 A. What happens if current is reversed without changing magnitude?

$U = LI^2/2 = (0.30)(4)/2 = 0.60$ J. Since energy depends on I^2, reversing I changes field direction but not stored energy for an ideal linear inductor.

CHECK THE ENDPOINT

$U = 0.60$ J for either current direction.

Work it through · Q30

Skill: Work to establish current. Model · syllabus-based

Derive the stored energy of a constant-L inductor from the work supplied by a source.

1. The source balances passive inductor drop $v = L\,dI/dt$.
   Why: It must overcome the opposing induced emf.
2. Power is $p = vI = LI\,dI/dt$.
   Why: Power delivered equals voltage times current.
3. Energy increment is $dU = p\,dt = LI\,dI$.
   Why: Time cancels in the differential expression.
4. Integrate from zero to I: $U = L\int_0^I I'\,dI' = LI^2/2$.
   Why: L is constant and initial field energy is zero.

CHECK THE ENDPOINT

$U = LI^2/2$ for a linear inductor.

Work it through · Q31

Skill: Field energy density calculation. Model · syllabus-based

Find the vacuum magnetic energy density for B=0.020 T, and the energy in 0.010 m³ of uniform field.

1. $u_B = B^2/(2\mu_0)$.
   Why: The region is vacuum, so $H = B/\mu_0$.
2. $u_B = 0.0004/(8\pi \times 10^{-7}) = 159.2$ J/m³.
   Why: Square B before dividing.
3. $U = u_B V = 1.592$ J.
   Why: The energy density is uniform.

CHECK THE ENDPOINT

$u_B = 159.2$ J/m³; U=1.592 J.

Work it through · Q39

Skill: Circuit-to-field magnetic energy. Model · syllabus-based

Derive magnetic energy from circuit work and recover the energy density inside a long linear solenoid. Discuss assumptions.

1. The incremental source work is $dU = LI\,dI$ for constant L.
   Why: The passive voltage is $L\dot{I}$.
2. Integration gives $U = LI^2/2$.
   Why: Start from zero current.

3. A long solenoid has $L = \mu N^2 A/\ell$ and $H = NI/\ell$.
   Why: Use its uniform interior field, neglecting ends.

4. Substitute to get $U = (\mu H^2/2)A\ell$.
   Why: Replace current and turns by H.

5. Divide by field volume $A\ell$ to get $u_B = \mu H^2/2 = B \cdot H/2 = B^2/(2\mu)$.
   Why: The medium is linear and isotropic.

6. In vacuum use μ_0. For nonlinear or hysteretic media, energy changes require the appropriate integral of $H \cdot dB$ and can include loss.
   Why: A constant proportionality no longer describes the material path.

CHECK THE ENDPOINT

In a linear lossless medium, $u_B = B \cdot H/2$; in vacuum, $u_B = B^2/(2\mu_0)$.

T8 Charge conservation and Maxwell's repair

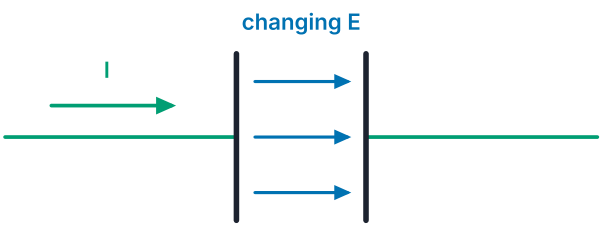
> Purpose: Ampère's original law cannot describe charge building up on a capacitor.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Ampère's original law cannot describe charge building up on a capacitor.



Gap: displacement contribution

During charging, conduction current in the leads and changing electric flux in the gap give equal total current contributions.

1. Divergence of a curl is zero, so the old law would require $\nabla \cdot J = 0$ always.
   Why: Apply divergence to both sides.
2. Continuity instead gives $\nabla \cdot J = -\partial\rho/\partial t$.
   Why: Charge can accumulate locally.
3. Gauss's law turns $\partial\rho/\partial t$ into $\epsilon_0 \nabla \cdot \partial E/\partial t$.
   Why: The changing charge changes electric divergence.

4. Add the displacement-current term to make the total divergence zero.
   Why: This restores consistency with conservation.

RESULT

$\nabla \cdot \mathbf{J} = -\partial\rho/\partial t$; $\nabla \times \mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\partial\mathbf{E}/\partial t$ in vacuum.

EXAM TRAP

Displacement current is not necessarily charge transport through the dielectric gap.

CHECK YOURSELF

What crosses an ideal vacuum capacitor gap: conduction current or displacement-current contribution?
Answer: Displacement-current contribution.

This topic supports Q15, Q16, Q32, Q37.

Work it through · Q15

Skill: Continuity equation. Model · syllabus-based

Write and interpret the local continuity equation.
$\partial\rho/\partial t + \nabla \cdot J = 0$. A positive net outward current from a tiny volume reduces its charge density. This is local conservation of electric charge.

CHECK THE ENDPOINT

Charge accumulation balances net inward current.

Work it through · Q16

Skill: Displacement-current distinction. Model · syllabus-based

What is displacement current in vacuum? Does it require charge carriers crossing the surface?

Its density is $J_d = \epsilon_0 \partial E/\partial t$, and its integrated contribution is $I_d = \epsilon_0 d\Phi_E/dt$. It can occur through a vacuum capacitor gap with no conduction carriers crossing.

CHECK THE ENDPOINT

Displacement current is a changing-field contribution to magnetic circulation.

Work it through · Q32

Skill: Charging capacitor displacement current. Model · syllabus-based

An ideal parallel-plate vacuum capacitor has area A and charging current I. Show that the displacement current through the gap equals I.

1. $E = Q/(\epsilon_0 A)$ between the plates away from edges.
   Why: Gauss's law gives the uniform field.
2. Electric flux through the plate area is $\Phi_E = EA = Q/\epsilon_0$.
   Why: Use the same area that the conduction-current loop spans.
3. $I_d = \epsilon_0 d\Phi_E/dt = dQ/dt = I$.
   Why: The change in charge creates the change in flux.

4. Thus a surface cutting the wire and a surface bulging through the gap give the same total Ampère-Maxwell contribution.
   Why: The extra term removes surface-choice inconsistency.

CHECK THE ENDPOINT

$I_d = I$ for the complete plate area.

Work it through · Q37

Skill: Derivation of Maxwell correction.

Legacy PYQ N25 Q21 · expanded

Define line, surface and volume current measures, derive charge continuity, then explain Maxwell's correction to Ampère's law using a charging capacitor.

1. For line charge density λ moving along a wire at speed v, $I = \lambda v$ in A. Surface current density is $K = \sigma_q v$ in A/m, and volume current density is $J = \rho v$ in A/m² for a single moving charge population.
   Why: Each is charge density times velocity, with the appropriate cross-section dimension.

2. Charge leaving a fixed volume per time is $\oint J \cdot da = -d \int \rho \, d\tau / dt$.
   Why: Outward flow reduces the charge remaining inside.

3. The divergence theorem converts this to $\int(\nabla \cdot J + \partial_t \rho) d\tau = 0$. Since every volume satisfies it, $\nabla \cdot J = -\partial_t \rho$.
   Why: A continuous integrand whose integral vanishes in every volume must vanish locally.

4. The old law is $\nabla \times B = \mu_0 J$. Taking its divergence forces $\nabla \cdot J = 0$.
   Why: Divergence of curl vanishes identically.

5. Continuity demands $\nabla \cdot J = -\partial_t \rho$, which is nonzero where charge accumulates.
   Why: Local charge conservation must hold.

6. Gauss's law gives $\rho = \epsilon_0 \nabla \cdot E$, so $\partial_t \rho = \epsilon_0 \nabla \cdot \partial_t E$.
   Why: Differentiate the electric source law.

7. Adding $\epsilon_0 \partial_t E$ to J makes the total divergence zero.
   Why: The two divergence terms cancel.

8. The corrected law is $\nabla \times B = \mu_0 (J + \epsilon_0 \partial_t E)$.
   Why: It contains conduction and displacement-current density.

9. For a vacuum capacitor $E = Q/(\epsilon_0 A)$, giving $I_d = \epsilon_0 A \dot{E} = \dot{Q} = I$.
   Why: Both possible spanning surfaces now give identical magnetic circulation.

CHECK THE ENDPOINT

Maxwell's displacement current restores charge conservation and surface independence.

T9 The four Maxwell equations

Purpose: Four field laws connect electric and magnetic sources and changes.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Four field laws connect electric and magnetic sources and changes.

1. Electric charge sources electric flux.
   Why: This is Gauss's electric law.
2. Magnetic flux through a closed surface is zero.
   Why: There are no magnetic monopoles in this formulation.
3. Changing magnetic field creates electric circulation.
   Why: This is Faraday's law.
4. Current and changing electric field create magnetic circulation.
   Why: This is Ampère-Maxwell law.

RESULT

$\nabla \cdot E = \rho/\epsilon_0$; $\nabla \cdot B = 0$; $\nabla \times E = -\partial_t B$; $\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \partial_t E$.

EXAM TRAP

The vacuum equations use total charge/current. In matter, switching to free sources also requires D and H.

CHECK YOURSELF

Which equation forbids a net magnetic flux from a closed surface?
Answer: $\nabla \cdot B = 0$.

This topic supports Q17, Q18, Q33.

Work it through · Q17

Skill: Gauss magnetic law.

Prerequisite-course PYQ A26 Q15 · adapted marks

Show that the divergence of a magnetic field vanishes, and state its integral form.

$\oint B \cdot da = 0$ and $\nabla \cdot B = 0$. Writing $B = \nabla \times A$ gives $\nabla \cdot B = \nabla \cdot (\nabla \times A) = 0$ because mixed derivatives cancel for a smooth vector potential. Thus magnetic field lines have no net source or sink: as much flux enters a closed surface as leaves it.

CHECK THE ENDPOINT

No magnetic monopole source appears in these equations.

Work it through · Q18

Skill: Faraday differential law. Model · syllabus-based

Write Faraday's law in differential form and explain each side.

$\nabla \times E = -\partial B/\partial t$. The left side measures local electric circulation per area; the right side gives the opposing rate of magnetic-field change.

CHECK THE ENDPOINT

Changing B generates circulating E.

Work it through · Q33

Skill: Integral Maxwell equations.

Legacy PYQ N25 Q6 · expanded to Maxwell system

Write the four integral Maxwell equations for a fixed loop/surface in vacuum, explaining the source or change in each.

1. $\oint_S E \cdot da = Q_{\text{enc}}/\epsilon_0$.
   Why: Enclosed electric charge produces electric flux.
2. $\oint_S B \cdot da = 0$.
   Why: No magnetic charge sources net closed-surface flux.
3. $\oint_C E \cdot dl = -d\int_S B \cdot da/dt$.
   Why: Changing magnetic flux produces electric circulation.
4. $\oint_C B \cdot dl = \mu_0 I_{\text{enc}} + \mu_0\epsilon_0 d\int_S E \cdot da/dt$.
   Why: Both conduction and changing electric flux produce magnetic circulation.

5. In magnetostatics the displacement term vanishes, leaving $\oint B \cdot dl = \mu_0 I_{\text{enc}}$, the original integral Ampère law.
   Why: All fields are time-independent.

6. For curl laws, C is the boundary of S and their orientations obey the right-hand rule.
   Why: A consistent sign requires paired path and surface directions.

CHECK THE ENDPOINT

The two flux laws and two circulation laws form Maxwell's system.

T10 Maxwell equations in matter

Purpose: D and H absorb the bound-charge and magnetization contributions.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

D and H absorb the bound-charge and magnetization contributions.

1. Separate total charge into free and bound charge.
   Why: P represents the bound electric response.
2. Separate current into free, magnetization and polarization currents.
   Why: Changing P is also a current contribution.
3. Use $D = \epsilon_0 E + P$ and $H = B/\mu_0 - M$.
   Why: These definitions collect the material terms.
4. Supply constitutive relations to solve a particular medium.
   Why: Four differential laws alone do not specify its response.

RESULT

$\nabla \cdot D = \rho_f$, $\nabla \cdot B = 0$, $\nabla \times E = -\partial_t B$, $\nabla \times H = J_f + \partial_t D$.

EXAM TRAP

$\partial_t D$ includes both vacuum and polarization terms; it is not always just $\epsilon_0 \partial_t E$.

CHECK YOURSELF

In a homogeneous linear dielectric, what is the displacement-current density?
Answer: $\epsilon\, \partial E/\partial t$.

This topic supports Q19, Q34.

Work it through · Q19

Skill: Free-source Maxwell law. Model · syllabus-based

State Ampère-Maxwell law in matter and identify the sources on its right side.

$\nabla \times H = J_f + \partial D/\partial t$. The first term is free conduction/current density; the second includes vacuum and polarization response because $D = \epsilon_0 E + P$.

CHECK THE ENDPOINT

The H curl is driven by free current plus changing D.

Work it through · Q34

Skill: Polarization current. Model · syllabus-based

Show how changing polarization contributes to current and enters Ampère-Maxwell law in matter.

1. A changing dipole density means charges shift locally, giving $J_p = \partial P/\partial t$.
   Why: Current includes bound charge motion.
2. Total current is $J = J_f + \nabla \times M + \partial P/\partial t$.
   Why: Magnetization loops and polarization changes are both bound contributions.
3. Insert this into vacuum-form Ampère-Maxwell and move curl M left.
   Why: Separate the material current terms.
4. The result is $\nabla \times H = J_f + \partial(\epsilon_0 E + P)/\partial t$.
   Why: Use the definition of H.

CHECK THE ENDPOINT

$\nabla \times H = J_f + \partial D/\partial t.$

T11 Electromagnetic boundary conditions

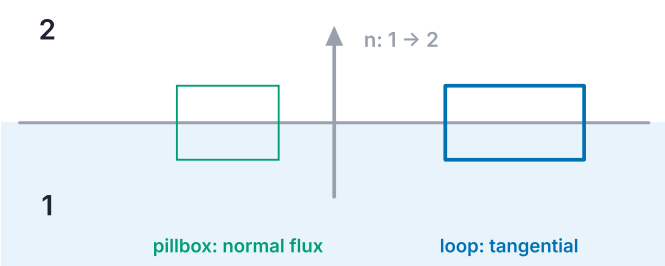
Purpose: Surface charges and currents determine the allowed field jumps.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. Current is charge per second. Voltage is energy per charge. A changing quantity has a rate of change; flux also depends on area and orientation.

THE IDEA

Surface charges and currents determine the allowed field jumps.



A pillbox compares normal fields. A shrinking loop compares tangential fields. Keep the same normal direction in every jump.

1. Use thin pillboxes for D and B.
   Why: Only the normal cap flux survives.
2. Use shrinking loops for E and H.
   Why: Only tangential circulation and any sheet current survive.
3. Keep n directed from medium 1 to 2 in all four relations.
   Why: A consistent orientation fixes the signs.
4. The formulas assume finite nonsingular fields at an ordinary stationary interface.
   Why: Flux derivative terms vanish with the shrinking loop area.

RESULT

$\hat{n} \cdot (D_2 - D_1) = \sigma_f$, $\hat{n} \cdot (B_2 - B_1) = 0$; $\hat{n} \times (E_2 - E_1) = 0$; $\hat{n} \times (H_2 - H_1) = K_f$.

EXAM TRAP

Continuity of tangential H requires zero free surface current; continuity of normal D requires zero free surface charge.

CHECK YOURSELF

With $K_f = 0$, is tangential B always continuous?
Answer: No. H is continuous; B can change if permeability changes.

This topic supports Q20, Q35, Q36, Q40.

Work it through · Q20

Skill: Normal magnetic continuity. Model · syllabus-based

Why is normal B continuous at an ordinary interface, even if permeability changes?

A thin pillbox and $\oint B \cdot da = 0$ give $B_{2n} = B_{1n}$. Permeability affects the corresponding H components, but there is no magnetic surface charge in Maxwell's equations.

CHECK THE ENDPOINT

Normal B is continuous.

Work it through · Q35

Skill: Boundary jump from a sheet current. Model · syllabus-based

An interface has normal $\hat{n} = \hat{z}$ from medium 1 to 2 and free surface current $K_f = K\hat{x}$. What tangential H jump is required? Also state the normal B condition.

1. Write $\Delta H = H_2 - H_1$. The condition is $\hat{z} \times \Delta H = K\hat{x}$.
   Why: A narrow Amperian loop encloses the sheet current.

2. Since $\hat{z} \times \hat{y} = -\hat{x}$, $\Delta H_t = -K\hat{y}$.
   Why: Choose the component with the required cross product.

3. Normal B satisfies $B_{2z} = B_{1z}$.
   Why: A magnetic pillbox encloses no magnetic charge.

4. The normal H components require permeability information if the media differ.
   Why: B and H are not interchangeable.

CHECK THE ENDPOINT

$H_{2t} - H_{1t} = -K\hat{y}$; normal B is continuous.

Work it through · Q36

Skill: Uncharged current-free interface. Model · syllabus-based

At a stationary interface with no free surface charge/current, medium 2 has $\epsilon_2 = 2\epsilon_1$, $\mu_2 = 3\mu_1$. Relate all normal and tangential E and H components.

1. $D_{2n} = D_{1n}$ gives $E_{2n} = E_{1n}/2$.
   Why: No free surface charge is enclosed by a thin pillbox.

2. $E_{2t} = E_{1t}$.
   Why: Faraday flux through a vanishing loop tends to zero for finite fields.

3. $B_{2n} = B_{1n}$ gives $H_{2n} = H_{1n}/3$.
   Why: Normal magnetic flux is continuous.

4. $H_{2t} = H_{1t}$.
   Why: There is no free sheet current.

CHECK THE ENDPOINT

Normal E and H scale inversely with ε and μ; tangential E and H are continuous here.

Work it through · Q40

Skill: Derivation of all interface conditions. Model · syllabus-based

Derive the four macroscopic electromagnetic boundary conditions at a stationary ordinary interface carrying free surface charge σf and free surface current Kf.

1. Take n from medium 1 to 2 and shrink a pillbox across the surface.
   Why: This fixes the cap orientations and jump signs.
2. Gauss's law for D leaves $(D_2 - D_1) \cdot n = \sigma_f$.
   Why: Side flux vanishes, but enclosed sheet charge stays proportional to cap area.
3. Magnetic Gauss law gives $(B_2 - B_1) \cdot n = 0$.
   Why: There is no magnetic surface charge.
4. A tiny rectangular loop in Faraday's law gives $n \times (E_2 - E_1) = 0$.
   Why: The magnetic-flux derivative tends to zero with loop area for bounded fields.
5. The H circulation law gives $n \times (H_2 - H_1) = K_f$.
   Why: The free sheet current survives while the displacement flux through the narrow loop vanishes.
6. Then use material laws, such as $D_i = \epsilon_i E_i$ and $B_i = \mu_i H_i$, to obtain jumps of other components.
   Why: Constitutive relations differ across the interface.

CHECK THE ENDPOINT

Normal D jumps by σf, normal B is continuous, tangential E is continuous, and tangential H jumps according to Kf.

Formula and condition sheet

T1 · Ohm’s law in a material

$\mathbf{J} = \sigma \mathbf{E}$ for an ohmic conductor at rest; $R = \ell/(\sigma A)$.

T2 · Electromotive force

$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l}$ where f is driving force per unit charge.

T3 · Motional emf and energy supply

$\mathcal{E} = B\ell v$ for mutually perpendicular B, rod length and velocity.

T4 · Faraday’s law and Lenz’s rule

$\mathcal{E} = -d\Phi_B/dt$ for a loop; $\mathcal{E} = -N\, d\Phi_B/dt$ for N identical turns.

T5 · Induced electric field

$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$; $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$.

T6 · Self and mutual inductance

$N\Phi = LI$; $\mathcal{E}_L = -L\, dI/dt$ for constant L; $N_2\Phi_{21} = MI_1$.

T7 · Magnetic field energy

$U = LI^2/2$; in a linear nondispersive medium $u_B = \mathbf{B} \cdot \mathbf{H}/2$. In vacuum $u_B = B^2/(2\mu_0)$.

T8 · Charge conservation and Maxwell's repair

$\nabla \cdot \mathbf{J} = -\partial\rho/\partial t$; $\nabla \times \mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\partial\mathbf{E}/\partial t$ in vacuum.

T9 · The four Maxwell equations

$\nabla \cdot E = \rho/\epsilon_0$; $\nabla \cdot B = 0$; $\nabla \times E = -\partial_t B$; $\nabla \times B = \mu_0 J + \mu_0\epsilon_0\partial_t E$.

T10 · Maxwell equations in matter

$\nabla \cdot D = \rho_f$, $\nabla \cdot B = 0$, $\nabla \times E = -\partial_t B$, $\nabla \times H = J_f + \partial_t D$.

T11 · Electromagnetic boundary conditions

$\hat{n} \cdot (D_2 - D_1) = \sigma_f$, $\hat{n} \cdot (B_2 - B_1) = 0$; $\hat{n} \times (E_2 - E_1) = 0$; $\hat{n} \times (H_2 - H_1) = K_f$.

Past-paper route

Q17 → T9
Prerequisite-course PYQ A26 Q15 · adapted marks

Q33 → T9
Legacy PYQ N25 Q6 · expanded to Maxwell system

Q37 → T8
Legacy PYQ N25 Q21 · expanded